1) a) $20V = 4V + 8\Omega\, I \rightarrow I = 2A$

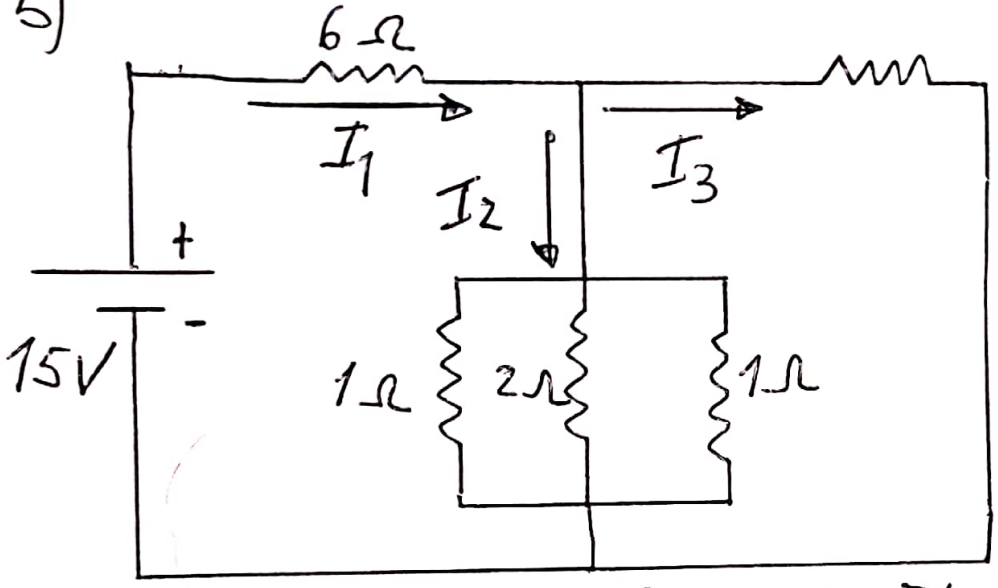
$4V = 2A\, R_{//} \rightarrow R_{//} = 2\Omega$

REEMPLAZANDO LA RESISTENCIA DE 8Ω POR 4Ω:

$R_{ef} = R_{//} + 4\Omega = 6\Omega \rightarrow I' = \frac{20V}{6\Omega} = 3{,}33\ A$

$P = VI' = 20V \times 3{,}33A = \boxed{66{,}7\ W}$

b)



$I_2 = 0{,}8A + 0{,}8A + \frac{0{,}8A}{2}$

$I_2 = 2A$

$\frac{1}{R_{//}} = \frac{1}{1\Omega} + \frac{1}{2\Omega} + \frac{1}{1\Omega}$

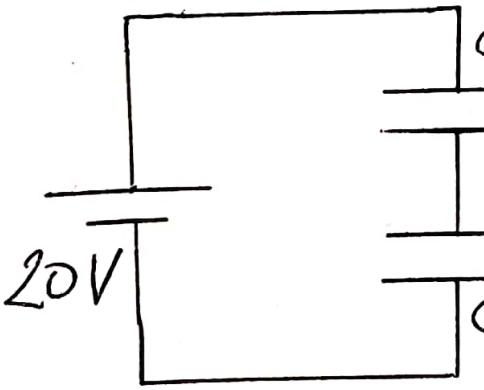
$R_{//} = 0{,}4\Omega$

$0 = 15V - 6\Omega I_1 - 0{,}4\Omega I_2 \rightarrow I_1 = \frac{71}{30}A = 2{,}37\ A$

$I_1 = I_2 + I_3 \rightarrow I_3 = \frac{11}{30}A = 0{,}37A$

$0 = -RI_3 + R_{//}I_2 \rightarrow R = \frac{I_2}{I_3}R_{//} = \frac{60}{11} \times 0{,}4\Omega = \boxed{2{,}18\ \Omega}$

2) CON L_1 CERRADA Y L_2 ABIERTA:



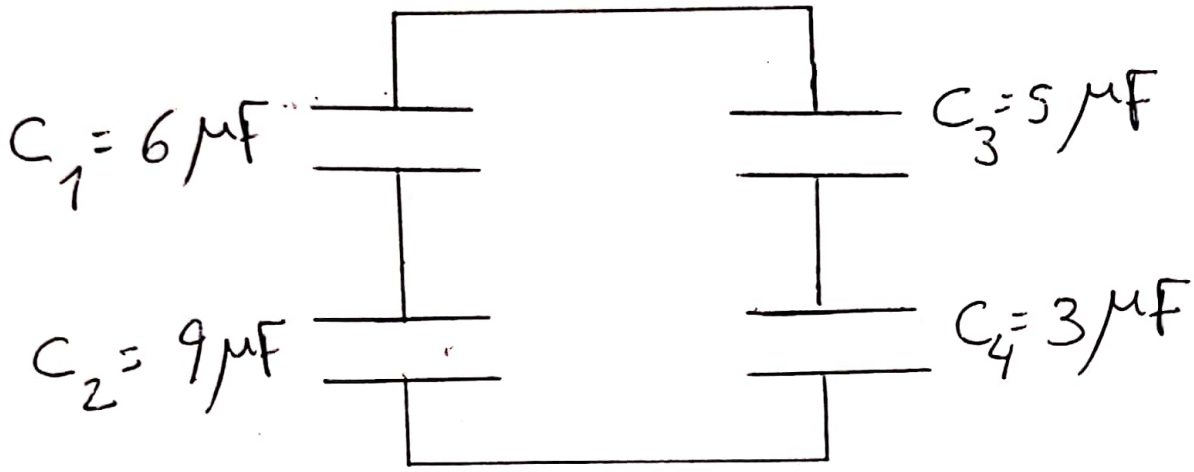
$Q_1 = Q_2$ (SERIE)

$0 = 20V - Q_1\left(\frac{1}{6\mu F} + \frac{1}{9\mu F}\right)$

$Q_1 = Q_2 = 72\ \mu C$

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CON L_1 ABIERTA Y L_2 CERRADA:



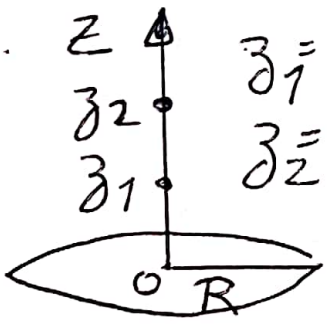
$Q_1' = Q_2'$ Y $Q_3' = Q_4'$ (SERIE)

$Q_1' + Q_3' = 72\,\mu C$

$$0 = Q_1'\left(\frac{1}{9\,\mu F} + \frac{1}{6\,\mu F}\right) - Q_3'\left(\frac{1}{5\,\mu F} + \frac{1}{3\,\mu F}\right) \rightarrow Q_3' = 0{,}52\,Q_1'$$

$$\boxed{\begin{aligned} Q_1' = Q_2' &= 47{,}4\,\mu C \\ Q_3' = Q_4' &= 24{,}6\,\mu C \end{aligned}}$$

3)



$z_1 = 12\,cm$

$z_2 = 18\,cm$

$$\vec{E} = \frac{kQz}{(R^2+z^2)^{3/2}}\hat{k} = \frac{2\pi k R \lambda z}{(R^2+z^2)^{3/2}}\hat{k}$$

(HAGO COINCIDIR EL EJE Z CON EL EJE DEL ANILLO)

a) $$\vec{F} = q\vec{E}(z_1) = -\frac{8\times10^{-6}\,C \times 2\pi \times 9\times10^{9}\,\frac{N m^2}{C^2} \times 0{,}06\,m \times 7\times10^{-6}\,C/m \times 0{,}12\,m}{[(0{,}06\,m)^2 + (0{,}12\,m)^2]^{3/2}}\hat{k}$$

$$\boxed{\vec{F} = -9{,}44\,N\,\hat{k}}$$

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b) $V(z) = \frac{kQ}{(R^2+z^2)^{1/2}} = \frac{2\pi k\lambda R}{(R^2+z^2)^{1/2}}$

$W_c = -q\Delta V = -q[V(z_2)-V(z_1)] = -q2\pi k\lambda R\left[\frac{1}{\sqrt{R^2+z_2^2}} - \frac{1}{\sqrt{R^2+z_1^2}}\right]$

$W_c = 8\times10^{-6}C \times 2\pi \times 9\times10^{9}\frac{Nm^2}{C^2} \times 7\times10^{-6}\frac{C}{m} \times 0{,}06m \times$

$\times\left[\frac{1}{\sqrt{(0{,}06m)^2+(0{,}18m)^2}} - \frac{1}{\sqrt{(0{,}06m)^2+(0{,}12m)^2}}\right]$

$\boxed{W_c = -0{,}41\,J}$

4) LA SUPERFICIE DE GAUSS ES UN CILINDRO / $\rho(r) = \rho_0 r$

1º $r<R$ (DENTRO DEL CILINDRO)

$2\pi r h E_1 = \frac{\rho_0}{\varepsilon_0}\int_0^h dz \int_0^{2\pi} d\theta \int_0^r r\,r\,dr = 2\pi h \frac{\rho_0}{\varepsilon_0}\frac{r^3}{3} \rightarrow \vec{E}_1 = \frac{\rho_0}{3\varepsilon_0} r^2 \hat{r}$

2º $r>R$ (FUERA DEL CILINDRO)

$2\pi r h E_2 = 2\pi h \frac{\rho_0}{3}\frac{R^3}{3} \rightarrow \vec{E}_2 = \frac{\rho_0 R^3}{3\varepsilon_0}\frac{1}{r}\hat{r}$

a) $\vec{E}(4\,cm) = \vec{E}_1(4\,cm) = \boxed{5423{,}73\frac{V}{m}\hat{r}}$

b) $\vec{E}(12\,cm) = \vec{E}_2(12\,cm) = \boxed{9689{,}27\frac{V}{m}\hat{r}}$

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5) a) $R_1 = \frac{0{,}2\,m}{kA}$; $R_2 = \frac{0{,}4\,m}{3kA} = \frac{2}{3}R_1$; $R_{12} = R_1 + R_2 = \frac{5}{3}R_1$

$I = \frac{\Delta T}{R_{12}} = 44^\circ C \times \frac{3}{5R_1} = \frac{26{,}4^\circ C}{R_1}$

$T - 5^\circ C = R_1 I = 26{,}4^\circ C \longrightarrow \boxed{T = 31{,}4^\circ C}$

b) $\frac{R_{12}}{4} = R' = \frac{0{,}6\,m}{KA} \longrightarrow \frac{0{,}6\,m}{KA} = \frac{1}{4}\,\frac{5}{3}\,\frac{0{,}2\,m}{230\,\frac{W}{m^\circ C}\,A} \longrightarrow \boxed{K = 1656\,\frac{W}{m^\circ C}}$

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